TEXORA · LECTURE SHEET

Thermodynamics

The laws governing heat, work and internal energy: the first law and its application to the standard processes, heat capacities and Mayer's relation, adiabatic change, the second law, and the Carnot, Otto and refrigeration cycles with their efficiencies.

PHY-1203 · PHYSICS-II · TOPIC 4 · LEVEL 1 TERM 2

1. The first law

Thermodynamics studies systems through a few state variables (p, V, T) and the two ways energy crosses a boundary: **heat** Q (energy transferred because of a temperature difference) and **work** W (energy transferred by a moving boundary or shaft). The difference is stored as **internal energy** U — a state function, unlike Q and W. The **first law** (energy conservation) reads, with W the work done *by* the system:

$$\Delta U = Q - W, \quad W = \int p \, dV$$

Sign discipline solves most textbook problems: heat *into* the system is +, work done *by* the system is +, and ΔU follows. Around any complete cycle ΔU = 0, so net work out equals net heat in — the ledger of every engine.

2. The standard processes

PROCESS	CONDITION	WORK	FIRST-LAW RESULT
Isobaric	p constant	$W = p\Delta V$	$\Delta U = Q - p\Delta V$
Isochoric	V constant	$W = 0$	$\Delta U = Q$ (defines C_V)
Isothermal (ideal gas)	T constant → $\Delta U = 0$	$W = nRT \ln(V_2/V_1)$	$Q = W$ — all heat in becomes work out
Adiabatic	$Q = 0$; $pV^{\gamma} = \text{const}$, $TV^{\gamma-1} = \text{const}$	$W = (p_1V_1 - p_2V_2)/(\gamma - 1)$	$\Delta U = -W$ — work is paid from internal energy

WORKED EXAMPLES

Isothermal: 1 mol at 300 K doubles its volume: $W = 8.314 \times 300 \times \ln 2$ = **1.73 kJ**, supplied entirely as heat.

> Adiabatic: air (γ = 1.4) compressed to 1/8 of its volume from 300 K: $T_2 = 300 \times 8^{0.4} = 300 \times 2.30 \approx$ **689 K** — hot enough to ignite diesel fuel sprayed into it, which is exactly how a diesel engine works: no spark plug, just adiabatic compression.

Heat capacities measure the heat for a 1 K rise at constant volume (C_V) or pressure (C_p). At constant pressure the system must also push the boundary, so C_p > C_V; for an ideal gas **Mayer's relation** gives the difference exactly: C_p − C_V = R per mole. Combined with kinetic theory (C_V = fR/2), this explains the measured capacities of all simple gases (Topic 3). **Enthalpy** H = U + pV packages the constant-pressure case: ΔH = Q at constant p — the chemist's favourite, and the quantity behind heats of reaction in CHE-1205.

3. The second law

The first law permits absurdities (a cup of tea powering a ship by cooling the ocean) that never happen. The **second law** forbids them, in two classic forms:

- **Kelvin:** no cyclic engine can convert heat from a single reservoir entirely into work — some heat must be rejected to a colder reservoir.
- **Clausius:** heat cannot flow from cold to hot by itself; moving it that way (refrigeration) costs work.

The two statements are equivalent (violating one designs a machine that violates the other). Their quantitative measure — entropy — is Topic 5.

4. Cycles and efficiencies

The **Carnot cycle** (isothermal expansion at T_h, adiabatic expansion, isothermal compression at T_c, adiabatic compression) is the most efficient cycle possible between two temperatures; its efficiency depends on temperatures alone:

$$\eta_{Carnot} = 1 - T_c/T_h \quad \text{(temperatures in kelvin)}$$

Between 600 K and 300 K, η = 50%: of 1,000 J absorbed, at most 500 J becomes work and 500 J must be rejected — and real engines, bound by irreversibility, do worse. The **Otto cycle** (the petrol engine's idealisation: two adiabatics + two isochores) reaches $\eta = 1 - r^{(1-\gamma)}$, where r is the compression ratio: at r = 8, γ = 1.4, η ≈ 56% ideal, ~25–30% real. Run backwards, a cycle is a **refrigerator**, judged by its coefficient of performance COP = Q_c/W (heat removed per unit work): a freezer between 260 K and 310 K has Carnot COP = 260/(310 − 260) = 5.2 — five joules pumped per joule paid, which is why heat pumps beat resistive heaters.

5. Reversible and irreversible change

A **reversible** process is a sequence of equilibrium states, drivable backwards by an infinitesimal nudge (slow, frictionless, near-isothermal compression is the ideal). Real processes are **irreversible**: free expansion into a vacuum, friction, heat flowing across a finite temperature gap. In free expansion of an ideal gas Q = W = 0, so ΔU = 0 and the temperature is unchanged — yet the gas will never spontaneously return to half its volume. The first law sees nothing special; the second law does. Measuring that one-wayness is entropy's job.

6. Key points

- First law: ΔU = Q − W; U is a state function, Q and W are path quantities.
- Isothermal work nRT ln(V_2/V_1); adiabatic work paid from ΔU with pV^γ and TV^(γ−1) constant.
- C_p − C_V = R (Mayer); ΔH = Q at constant pressure.
- Second law (Kelvin and Clausius forms are equivalent): single-reservoir engines and free heat pumping are impossible.
- Carnot η = 1 − T_c/T_h caps every engine between the same temperatures; Otto η = 1 − r^(1−γ); refrigerator COP = Q_c/W.

7. Practice

Answer in your own words; brief model answers follow.

1. A system absorbs 500 J of heat and does 200 J of work. Find ΔU. It then returns to its start by a path on which 100 J of work is done on it; how much heat leaves in the return leg?
2. Two moles of ideal gas expand isothermally at 400 K from 10 L to 30 L. Find the work done and the heat absorbed.
3. Air at 300 K is compressed adiabatically to one tenth of its volume (γ = 1.4). Find the final temperature, and name the machine that exploits this.
4. A Carnot engine works between 500 K and 300 K, absorbing 2,000 J per cycle. Find its efficiency, work output and heat rejected.
5. An Otto engine has compression ratio 10 and γ = 1.4. Find its ideal efficiency and one reason the real engine falls short.
6. In the free expansion of an ideal gas into a vacuum, Q = W = ΔU = 0, yet the process cannot run backwards by itself. Which law points at the missing concept, and what is the concept?

Answer key

1. ΔU = Q − W = 500 − 200 = +300 J. Around the full cycle ΔU_total = 0, so the return leg has ΔU = −300 J; with W = −100 J (work done on the system): −300 = Q − (−100) → Q = −400 J: 400 J of

heat leaves.

2. W = nRT ln(V_2/V_1) = 2 × 8.314 × 400 × ln 3 = 6,651 × 1.099 ≈ 7.3 kJ. Isothermal ideal gas: ΔU = 0, so Q = W = 7.3 kJ absorbed.
3. $T_2 = T_1(V_1/V_2)^{(\gamma-1)}$ = 300 × 10^0.4 = 300 × 2.512 ≈ 754 K. The diesel engine — compression ignition with no spark.
4. η = 1 − 300/500 = 0.40 (40%). W = 0.40 × 2,000 = 800 J. Heat rejected Q_c = 2,000 − 800 = 1,200 J.
5. η = 1 − r^(1−γ) = 1 − 10^(−0.4) = 1 − 0.398 ≈ 60%. Real losses: heat loss to cylinder walls, friction, finite combustion time and incomplete combustion — all irreversibilities the ideal cycle omits.
6. The second law; the missing concept is entropy. The expanded state has higher entropy (more accessible microstates), and the total entropy of the universe cannot decrease — the one-way arrow the first law cannot see. Topic 5 quantifies it.
